

# Hydration and Nutrition for Recovery

Reference notes on general hydration and nutrition principles that support recovery between training sessions. General wellness information, not individualized dietary or medical advice.

## Hydration basics

Adequate fluid intake supports blood volume, temperature regulation, and nutrient transport, all of which affect recovery and next-day training performance. Simple practical indicators of hydration status include urine color (pale straw is a reasonable target) and unexplained day-to-day body-weight swings around training.

## Fluid needs around exercise

- General daily fluid needs vary by body size, climate, and activity level, so there is no single number that fits everyone.
- Replacing fluid lost during longer or sweatier sessions, and continuing to rehydrate afterward, supports faster recovery.
- In hot conditions or with heavy sweating, replacing electrolytes (particularly sodium) alongside water becomes more important.

## Nutrition principles that support recovery

- Protein intake spread across the day supports muscle repair after training.
- Carbohydrate intake after harder sessions helps replenish muscle glycogen stores, which supports readiness for the next session.
- Under-eating relative to training load over multiple days can itself suppress recovery markers such as HRV and sleep quality.
- Alcohol close to bedtime can reduce sleep quality even if it seems to help someone fall asleep faster.

## How this fits into a recovery assistant

Because this assistant works from wearable and training data rather than detailed food logs, nutrition guidance should stay general and directional (for example, noting that consistent under-fueling or poor hydration can compound a low recovery score) rather than prescribing specific calorie or macronutrient targets, which should come from a qualified nutrition professional.